

Intrusion Detection/ Prevention System and network scanners

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Intrusion detection

- Intrusion Detection
  - Intrusion detection is a technique of detecting unauthorized access to a computer system or a computer network.
  - An intrusion into a system is an attempt by an outsider to illegally gain access to the system.
  - **Intrusion prevention**, on the other hand, is the art of preventing an unauthorized access of a system's resources.
  - The two processes are related in a sense that while intrusion detection passively detects system intrusions, intrusion prevention actively filters network traffic to prevent intrusion attempts.

Intrusion

- An intrusion is a deliberate unauthorized attempt, successful or not, to break into, access, manipulate, or misuse some valuable property and where the misuse may result into or render the property unreliable or unusable.
- The person who intrudes is **an intruder.**

Types of intrusion

Attempted break-ins
which are detected by atypical behavior profiles or violations of security constraints.

Masquerade attacks
which are detected by atypical behavior profiles or violations of security constraints. These intrusions are also detected using anomaly-based IDS.

Penetrations of the security control system
which are detected by monitoring for specific patterns of activity.

Leakage
which is detected by atypical use of system resources.

Denial of service
which is detected by atypical use of system resources.

Malicious use
which is detected by atypical behavior profiles, violations of security constraints, or use of special privileges.

An intrusion detection system (IDS)

- An intrusion detection system (IDS) is a system used to detect unauthorized intrusions into computer systems and networks. Intrusion detection as a technology is not new, it has been used for generations to defend valuable resources.
- These are three models of intrusion detection mechanisms:
  - **anomaly-based detection**
  - **signature-based detection**
  - **hybrid detection.**

An intrusion detection system (IDS)

Anomaly Based IDS

- Anomaly based systems are "learning" systems in a sense that they work by continuously creating "norms" of activities. These norms are then later used to detect anomalies that might indicate an intrusion.
- Anomaly detection compares observed activity against expected normal usage.
- With an anomaly-based IDS, anything that does not align with the existing normalized baseline—such as a user trying to log in outside of standard business hours, new devices being added to a network without authorization, or a flood of new IP addresses trying to establish a connection with a network—will raise a potential flag for concern.

An intrusion detection system (IDS)

Signature Based IDS

- Signature-based IDS refers to the detection of attacks by looking for specific patterns, such as byte sequences in network traffic, or known malicious instruction sequences used by malware.
- It operates by using a pre-programmed list of known threats and their indicators of compromise (IOCs).
- An IOC might be a specific behavior that generally precedes a malicious network attack, file hashes, malicious domains, known byte sequences, or even the content of email subject headings.

An intrusion detection system (IDS)

Hybrid Detection

- Because of the difficulties with both the anomaly-based and signature-based detections, a hybrid model is being developed.
- Seeks to bring combination of signature and anomaly based detection into one.
- Much research is now focusing on this hybrid model.

Types of Intrusion detection

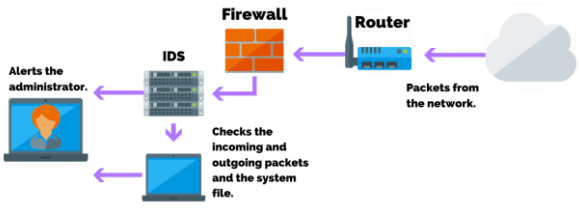
- Intrusion detection systems are classified based on their monitoring scope.
- There are: **network-based intrusion detection** and **host-based detections.**
- **Network-Based Intrusion Detection Systems (NIDSs)**
  - NIDSs have the whole network as the monitoring scope. They monitor the traffic on the network to detect intrusions.
  - As such, a typical NIDS has to include a packet sniffer to gather network traffic for analysis.
  - They are responsible for detecting anomalous, inappropriate, or other data that may be considered unauthorized and harmful occurring on a network. There are striking differences between NIDS and firewalls.

Types of Intrusion detection

Host-Based Intrusion Detection Systems (HIDS)

- Host-based intrusion detection is the technique of detecting malicious activities on a single computer.
- Host-Based Intrusion Detection Systems (HIDS)
  - A host-based intrusion detection system, is therefore, deployed on a single target computer and it uses software that monitors operating system specific logs including system, event, and security logs on Windows systems and syslog in Unix environments to monitor sudden changes in these logs.
  - When a change is detected in any of these files, the HIDS compares the new log entry with its configured attack signatures to see if there is a match.
  - If a match is detected then this signals the presence of an illegitimate activity.

Host Intrusion Detection System (HIDS)

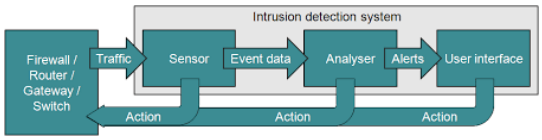


IDS Components

An IDS comprises of 3 logical components:

1. **Sensors:** Sensors are responsible for collecting data. The input for a sensor may be any part of a system that could contain evidence of an intrusion. Types of input to a sensor include network packets, log files, and system call traces. Sensors collect and forward this information to the analyzer.
2. **Analyzers:** Analyzers receive input from one or more sensors or from other analyzers. The analyzer is responsible for determining if an intrusion has occurred. The output may include evidence supporting the conclusion that an intrusion occurred. The analyzer may provide guidance about what actions to take as a result of the intrusion.
3. **User interface:** The user interface to an IDS enables a user to view output from the system or control the behavior of the system. In some systems, the user interface may equate to a manager, director, or console component.

IDS Components



Other Types of Intrusion Detection Systems

- Although NIDS and HIDS and their hybrids are the most widely used tools in network intrusion detection, there are others that are less used but more targeting and, therefore, more specialized.
- Because many of these tools are so specialized, many are still not considered as being intrusion detection systems but rather intrusion detection add-ons or tools.

Other types of IDS

- System Integrity Verifiers (SIVs)
- Honeypots

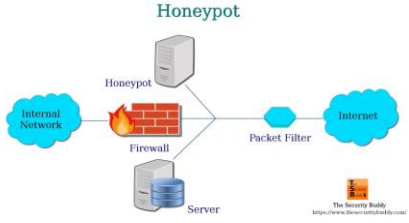
Other Types of IDS

- **System Integrity Verifiers (SIVs)**
  - SIVs monitor critical files in a system, such as system files, to find whether an intruder has changed them.
  - They can also detect other system components' data; for example, they detect when a normal user somehow acquires root/administrator level privileges.
  - In addition, they also monitor system registries in order to find well known signatures

Honeypot

- A honeypot is a system designed to look like something that an intruder can hack. They are built for many purposes but the overriding one is to deceive attackers and learn about their tools and methods.
- Honeypots are also add-on/tools that are not strictly sniffer-based intrusion detection systems like HIDS and NIDS.
- However, they are good deception systems that protect the network in much the same way as HIDS and NIDS.
- Since the goal for a honeypot is to deceive intruders and learn from them without compromising the security of the network, then it is important to find a strategic place for the honeypot. In the DMZ for those networks with DMZs or behind the network firewall if the private network does not have a DMZ.

Honeypot



Challenges of IDS

- Automated intrusion detection systems have a number of weaknesses.
  - They can be too sensitive, falsely reporting that an intrusion is under way, for example if a network is incorrectly configured or a buggy program begins issuing large numbers of packets.
  - Signature IDS relies on the software suppliers issuing regular updates to the list of known signatures, until the IDS receives the update it is effectively blind to the attack.

How to select IDS software for your network

- Selection, placement and maintenance of Intrusion detection (IDS) are mostly based on the requirements and current infrastructure of a company.
  - Understand that different types of intrusion detection tools do different jobs and which one would best fit the company- Do a proper research before choosing.
  - Understand company's requirements for IDS.
  - One product may work well for one company and fail for the next. That is why it is important to understand particular requirements and the unique infrastructures.

IDS Software's

IDS	HIDS/NIDS	Unix	Linux	Windows	Mac OS
SolarWinds Security Event Manager EDITOR'S CHOICE	Both	No	No	Yes	No
CrowdStrike Falcon (FREE TRIAL)	HIDS	Yes	Yes	Yes	Yes
Snort	NIDS	Yes	Yes	Yes	No
OSSEC	HIDS	Yes	Yes	Yes	Yes
ManageEngine EventLog Analyzer	HIDS	Yes	Yes	Yes	Yes
Suricata	NIDS	Yes	Yes	Yes	Yes
Zeek	NIDS	Yes	Yes	No	Yes
Sagan	Both	Yes	Yes	No	Yes
Security Onion	Both	No	Yes	No	No
AIDE	HIDS	Yes	Yes	No	Yes
Open WIPS-NG	NIDS	No	Yes	No	No
Samhain	HIDS	Yes	Yes	No	Yes

IDS Placement

- The most common placement being behind the firewall on the edge of a network.
- This practice provides the IDS with high visibility of traffic entering your network and will not receive any traffic between users on the network,the edge of the network is the point in which a network connects to the extranet.
- Sometimes an IDS with more advanced features will be integrated with a firewall in order to be able to intercept sophisticated attacks entering the network.
- Another option for IDS placement is within the actual network. These will reveal attacks or suspicious activity within the network.

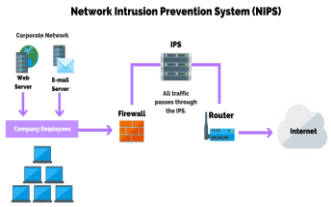
Intrusion Prevention Systems (IPSs)

- Although IDS have been one of the cornerstones of network security, they have covered only one component of the total network security picture since they have been and they are a passive component which only detects and reports without preventing.
- A promising new model of intrusion is developing and picking up momentum. It is the *intrusion prevention system* (IPS) which detects and prevent attacks.
- Like their counterparts the IDS, IPS fall into two categories:
  - Network-based IPS
  - Host-based IPS
- An IPS will generally be placed at an edge of the network, such as immediately inside an Internet firewall, or in front of a server farm

23

Intrusion Prevention Systems

An Intrusion Prevention System identifies suspicious activity, log security events, **attempt to block intrusions** and Report intrusion attempts.



Differences Between IDS and IPS

- The main difference between them is that IDS is a monitoring system, while IPS is a control system.
- IDS only issues alerts for potential attacks, while IPS can take action against them.
- Also, IDS is not inline, so traffic doesn't have to flow through it. Traffic does, however, have to flow through your IPS.
- False positives for IDS will only cause alerts, while false positives for IPS could cause the loss of important data or functions.

Differences Between IDS and IPS

- **An Intrusion Detection System (IDS)** is used to monitor a network, which then sends alerts when suspicious events on a system or network are detected.
- **An Intrusion Prevention System (IPS)** reacts to attacks in progress with the goal of preventing them from reaching targeted systems and networks.
- While IDS and IPS both have the ability to detect attacks, the main difference is in their responses to an attack.
- Intrusion Detection is passive- identifies threats and sends an alert. IDS requires a human or another system to look at the results and determine what actions to take next,
- Intrusion Prevention is active- Reacts to the attack. IPS works on its own to resolve the threats.

Response to System Intrusion

- A good intrusion detection system alert should produce a corresponding response.
- A good response must consist of pre-planned defensive measures that include an incident response team and ways to collect IDS logs for future use and for evidence when needed.
- It is therefore, important for Network Administrators to plan ahead of time how they will respond to system intrusion alerts.

Response to System Intrusion

- An *incident response team* (IRT) is a primary and centralized group of dedicated people charged with the responsibility of being the first contact team whenever an incidence occurs.
- An IRT must have the following responsibilities:
  - keeping up-to-date with the latest threats and incidents
  - being the main point of contact for incident reporting
  - notifying others whenever an incident occurs
  - assessing the damage and impact of every incident
  - finding out how to avoid exploitation of the same vulnerability
  - recovering from the incident.

Network Security Scanners

Network scanning

- Network scanning refers to a set of procedures used for identifying hosts, ports, and services in a network.
- Network scanning involves many procedures that help to identify the ports, services, and live hosts. It helps to discover the architecture and the operating system of the target system.
- Network scanning helps to find out the vulnerabilities and the threats in the articular network. Network changes also help to create a profile for the organization that is the target.
- Scanning involves collecting information making use of aggressive and complex techniques.

Network security scanner

- A network security scanner is a software tool that scans an entire network and its nodes for security vulnerabilities and loopholes. It is an automated solution that scans, assesses and evaluates the security posture and strength of the underlying network.
- A network security scanner is also known as a network vulnerability scanner.

Network security scanners

- Typically, a network security scanner scans all known and possible vulnerabilities and threats.
- It scans all network devices including:
  - Routers
  - Servers
  - Firewalls
  - Client computers

Network security scanners

- Network security scanners check for vulnerabilities such as:
  - Password strength
  - Open ports
  - Scripts
  - Operating system controls

Network security scanners

- Most scanners after analysis provide a reporting feature that reports:
  - IT assets
  - Associated vulnerabilities
  - Prioritized threats
  - Percentage of risk vulnerability

Network security scanners

- Regular scanning of your network allows you to keep track of the devices on your network, view how they're performing, spot the flaws, and understand the flow of traffic between connected devices and applications.
- It also helps to identify ,suspicious behaviors or traffic in a network which may be an indication that the network is under attack.

Types of Network Scanning

- Scanning is primarily of three types. These are:
  - Network scanning
  - port scanning
  - vulnerability scanning

Types of Network Scanning

A) Network scanning

- Network scanning helps to discover any live computer or hosts, open ports, and the IP address of a victim.
- It helps to discover the services that are running on any host computer.
- It allows the decoding of the system architecture of any target and the operating system.
- The method helps to deal with and discover if there are any vulnerabilities in a live host.

Types of Network Scanning

B) Port scanning

- Port scanning is a conventional method that is used to penetrate into the hackers and the testers to search if there are any open doors from where the hacker will be capable of accessing the system of the organization.
- It tries to figure out the route of the hacker, to find out the live hosts, the operating system that is used, and the installed firewalls as well as the topology of the targeted organization.
- Once the hacker gets the IP address of the organization of the victim using the UDP and the TCP ports the hacker will map the network of the organization and put it in his grab.
- A map is a tool that is used to carry out port scanning techniques.

Types of Network Scanning

C) Vulnerability scanning

- The vulnerability scanning method proactively identifies the vulnerability of the network in an automated method that helps to find out whether the system may be threatened or exploited.
- To carry out this type of scanning the computer needs to be connected to the internet.

Network Scanning Tools

1. Acunetix
2. SolarWinds Network Device Scanner
3. ManageEngine OpUtils
4. Intruder
5. Syxsense
6. PRTG Network Monitor
7. OpenVAS
8. Wireshark
9. Nikto
10. Angry IP Scanner
11. Advanced IP Scanner
12. Qualys Freescan
13. SoftPerfect Network Scanner
14. Retina Network Security Scanner
15. Nmap
16. Nessus
17. Metasploit Framework
18. Snort

https://www.softwaretestinghelp.com/network-scanning-tools/

